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CRACKING THE CODE OF LOAN
APPROVAL

Topic: Final project report

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Course: Introduction to Data Analysis

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I. INTRODUCTION

In the context of globalization and the rapid growth of the digital economy, the credit market in the United States and Canada has expanded significantly, with increasing demand for loans to support personal consumption, education, and business activities. This trend highlights the critical role of credit in driving investment, consumption, and overall economic growth, while also creating challenges for financial institutions that must balance accessibility with system stability. Loan approval processes are designed to manage this balance by collecting personal and financial information, reviewing credit history, and analyzing key ratios such as debt-to-income (DTI). However, rising levels of bad debt, unstable repayment capacity, and unpredictable market fluctuations underscore the need for modern data-driven approaches to strengthen decision-making and mitigate risk.

The objectives of this analysis are to identify the critical factors influencing loan approval decisions, evaluate the role of credit scores and loan intention in assessing borrower reliability and risk levels, detect potential biases in approval processes to ensure fairness and transparency, and strengthen credit risk management through data-driven insights that reduce exposure to bad debts and repayment instability. By applying modern data analytics methods, the study also aims to support more efficient and accurate decision-making, ultimately enhancing the resilience of the financial system.

II. EXPLORATORY DATA ANALYSIS

1. Data Overview

- **Source:** [Kaggle Datasets – Loan approval data 2025](#).
- **Coverage:** Customer profiles, credit history, and loan features in the United States and Canada.
- **Period:** First six months of 2025.
- **Size:** 50,000 rows and 20 columns.

2. Data Cleaning and Preprocessing

Missing Values: No null values were detected in the dataset.

Outlier Treatment: In this step, identifying the key variables helps us determine which factors have the greatest impact on the outcome.

df.isnull().sum()

	0	delinquencies_last_2yrs	0
customer_id	0	derogatory_marks	0
age	0	product_type	0
occupation_status	0	loan_intent	0
years_employed	0	loan_amount	0
annual_income	0	interest_rate	0
credit_score	0	debt_to_income_ratio	0
credit_history_years	0	loan_to_income_ratio	0
savings_assets	0	payment_to_income_ratio	0
current_debt	0	loan_status	0
defaults_on_file	0		

Figure 1.1. Summary of missing values per column

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id           50000 non-null  object
1   age                   50000 non-null  int64
2   occupation_status     50000 non-null  object
3   years_employed        50000 non-null  float64
4   annual_income         50000 non-null  int64
5   credit_score          50000 non-null  int64
6   credit_history_years   50000 non-null  float64
7   savings_assets        50000 non-null  int64
8   current_debt          50000 non-null  int64
9   defaults_on_file      50000 non-null  int64
10  delinquencies_last_2yrs 50000 non-null  int64
11  derogatory_marks       50000 non-null  int64
12  product_type           50000 non-null  object
13  loan_intent            50000 non-null  object
14  loan_amount            50000 non-null  int64
15  interest_rate          50000 non-null  float64
16  debt_to_income_ratio   50000 non-null  float64
17  loan_to_income_ratio   50000 non-null  float64
18  payment_to_income_ratio 50000 non-null  float64
19  loan_status            50000 non-null  int64
```

Figure 1.2. Variable types of dataset

From there, our group filtered this dataset using the above formula, resulting in 46,447 rows remaining.

Figure 2.3. Executing outliers removal based on IQR scale using Six Sigma methodology

Based on this, we proceeded to filter outliers in numeric features, except for certain specific columns such as loan_status, payment_to_income_ratio, debt_to_income_ratio, defaults_on_file, delinquencies_last_2yrs, and derogatory_marks. The method we used to filter outliers is Six Sigma, this is a quality management approach based on statistics, aimed at minimizing errors and variability in production processes or data analysis. The range of $\pm 6\sigma$ covers almost all data (99.99966%). Points outside this range have an extremely low probability (~ 3.4 defects per million opportunities).

Conversion to IQR scale:

$$(6\sigma) / \text{IQR} = 6 / 1.349 \approx 4.45 \text{ IQR}$$

$$\pm 6\sigma \approx \pm 4.45 \text{ IQR}$$

```
numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns.tolist()
numeric_cols = [c for c in numeric_cols if c != 'loan_status']

exclude_cols = [
    'loan_status',
    'payment_to_income_ratio',
    'debt_to_income_ratio',
    'defaults_on_file',
    'delinquencies_last_2yrs',
    'derogatory_marks'
]

numeric_cols1 = [c for c in numeric_cols if c not in exclude_cols]
for col in numeric_cols1:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 4.45 * IQR
    upper_bound = Q3 + 4.45 * IQR

    df = df[(df[col] >= lower_bound) & (df[col] <= upper_bound)]
df.shape
```

(46447, 20)

3. Variable Categorization

To facilitate analysis, key variables were categorized as follows:

- **The 'Credit Score' column according to the FICO scale:**
 - 300–579: Poor
 - 580–669: Fair
 - 670–739: Good
 - 740–799: Very Good
 - 800–850: Exceptional
- **The DTI ratio into three ranges:**
 - DTI below 36%
 - DTI from 36% to 49%
 - DTI above 50%

III. FINDINGS

1. The common features of declined loan offers

By analyzing the dataset, we will outline some common features of the declined applications so the customers can figure out how to improve the chance of getting the approval.

1.1. Violation record

We have identified the most noticeable figure of the loan application rejection which is the violation record, including data from the columns "delinquencies_last_2yrs", "derogatory_marks", and "defaults_on_file". To find out the relationship between them, we visualized through the charts.

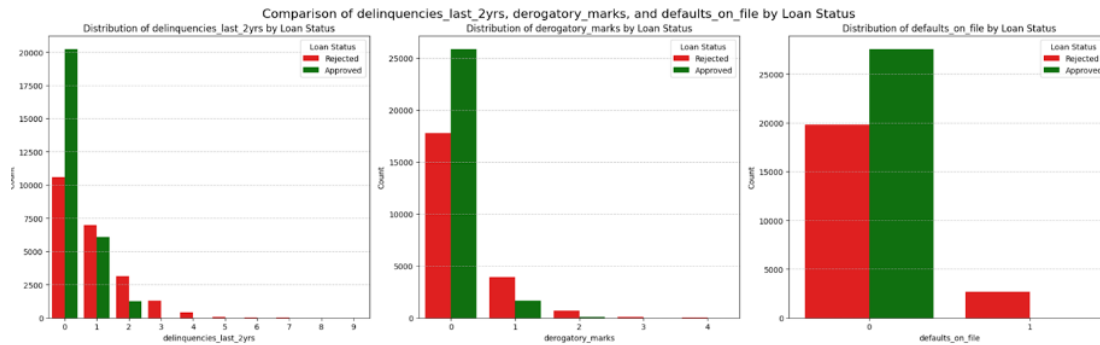


Figure 2.1.1. Charts of Approvals by Violation Category

As the number of violations increases, the rejection rate rises significantly. Customers with one default on file are never approved, and even borrowers with high credit scores and stable income face the same rejection outcome due to repeated contract violations.

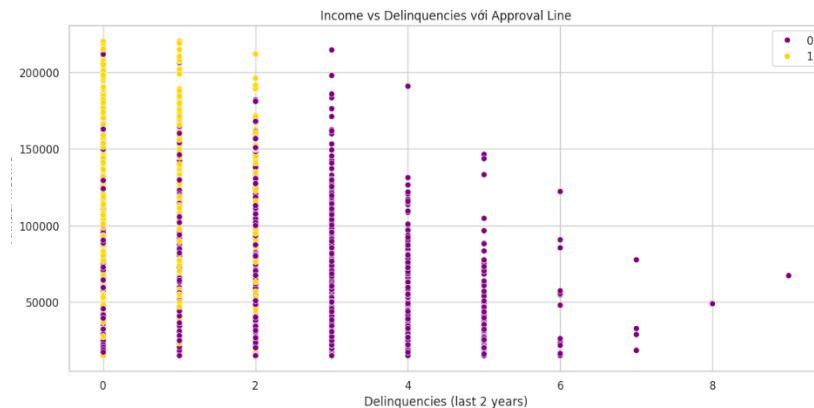
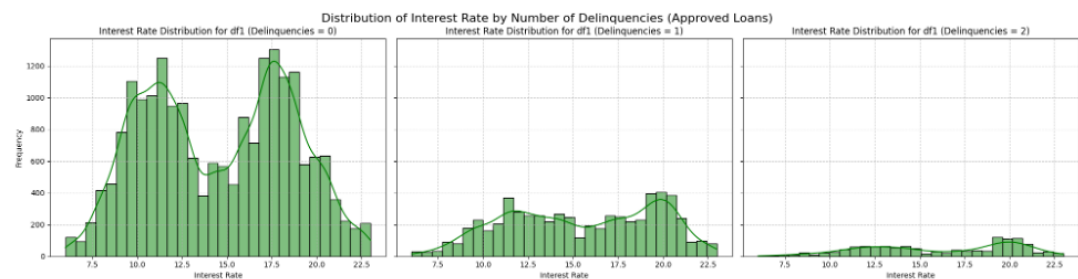


Figure 2.1.2. Scatterplot of delinquencies & income group

To examine whether loan amount and interest rate are affected by violation records, we plotted histograms of approved loans across different levels of delinquencies in the last two years and derogatory marks, using interest rate distributions as a representative indicator.

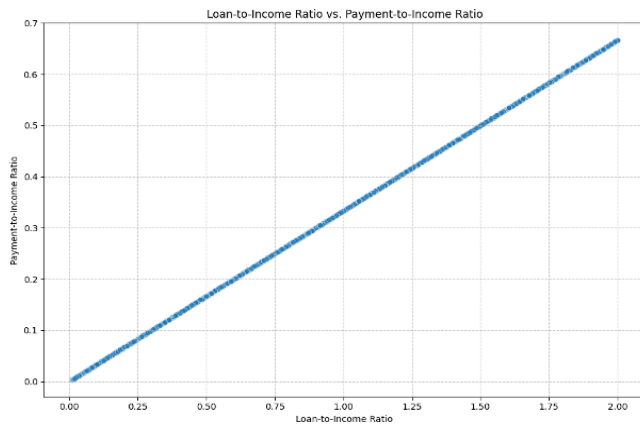
Figure 2.1.3. Interest Rate Distribution by Delinquencies (Last 2 Years)



From the above histograms, it is obvious that the shapes of the distributions remain the same while

the "delinquencies_last_2yrs", "derogatory_marks" change; only the frequencies go down. This proves that these factors are not influenced by the borrowers' violation history.

1.2. The in-time paying ability



Moving to LTI (loan_to_income_ratio) and PTI (payment_to_income_ratio), these figures are considered to represent the customers' ability to pay the loans in loan terms. Through analyzing those, we discover the fact that those two figures are linear variables due to the below scatterplot, so we would pick one out to be the common representative.

Figure 2.1.4 Scatterplot of "loan_amnt" "payment_to_income_ratio"

Figure 2.1.5 Area chart of rejected and approved loans by "loan_to_income_ratio"

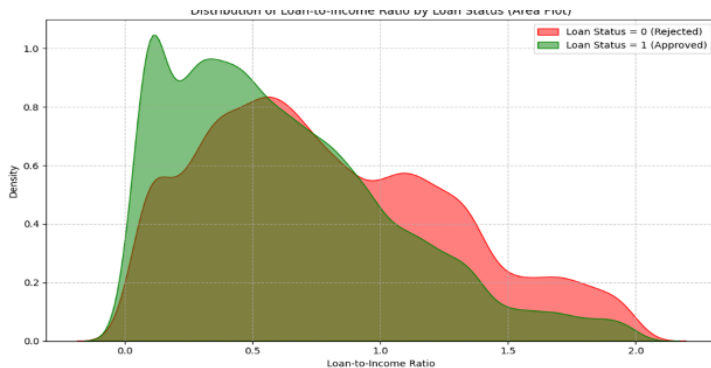
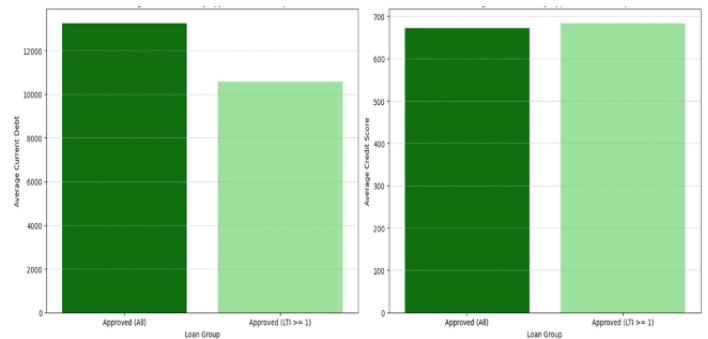


Figure 2.1.6 Bar chart comparing credit score and current debt, split between all approved loans and approved loans with LTI ≥ 1 .



The given area plot shows that if the LTI is lower than 0.5, the applications are more likely to get accepted. When the figure exceeds that point, the proportion of getting approval will experience a plummet. When the LTI is pretty high, the banks are likely to turn down the offers. Nonetheless, there are some offers that can still get approved though the LTI is high.

1.3. The payback potential

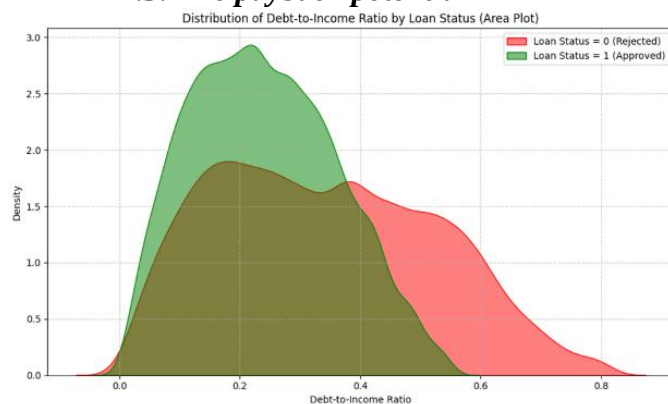


Figure 2.1.7. Area chart of rejected and approved loans by DTI

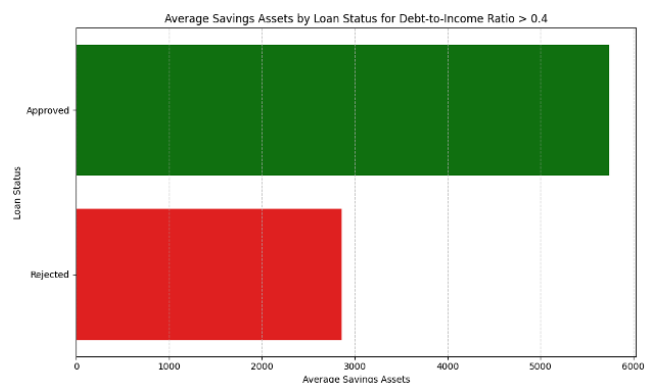


Figure 2.1.8. Bar chart of "savings_assets" for loans with a DTI > 0.4, separated into approved and rejected loans

Consequently, the DTI (debt_to_income_ratio) area plot shows that the banks prefer the offers with DTI lower than 0,4. When this figure is high, it is truly risky to make the offers because the banks conclude that the customers are not capable of carrying more debts. However, there are a few extraordinary cases where the banks still make the loans though the DTI is pretty high

If the DTI is high, the banks will evaluate the saving assets to have a clear overview.

1.4. Conclusion

When evaluating an offer, the banks will consider:

- **Violation record:** When having breached out the contracts many times, the percentages of getting the approval will decrease heavily. Especially, when the customer has ever gotten bankrupted, they will be unable to get the approval once again
- **The paying ability in time:** The banks would like to keep the DTI and PTI of the contracts low, like LTI smaller than 0.5. However, it can still be influenced by other figures like current debt or credit score
- **The payback potential:** The banks are not afraid of lending a lot of money, they just want to recover the capital. Although high DTI is a disadvantage, customers can demonstrate their finance ability through high saving assets to create a secure relationship with the banks

2. Key factors that increase the loan approval chance

2.1. Credit Score - Borrowers' trustworthiness

After analyzing the loan application dataset to determine the factors that most increase a borrower's chances of getting approved, we found that Credit Score (which represents character, the first C in the Five Cs of Credit) has the strongest relationship with loan approval outcomes.

Distributional characteristics of the Credit score

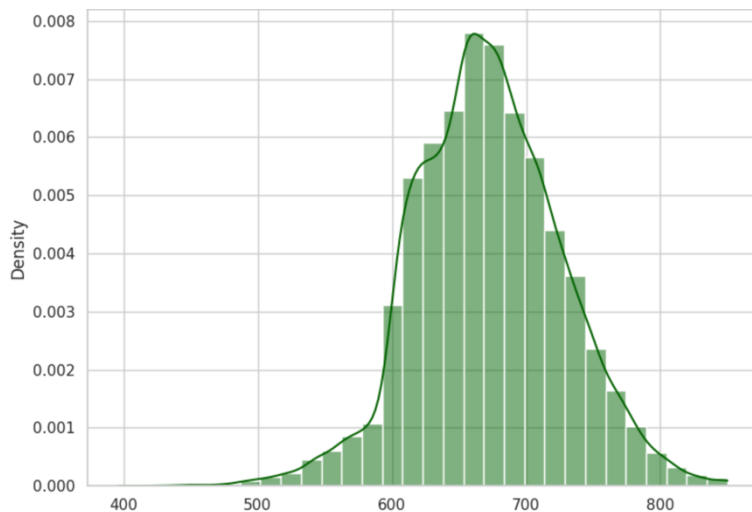


Figure 2.2.1. Credit score distribution among Approved loan applications

The dataset shows a left-skewed distribution of credit scores among approved applications, which indicates that most approvals clustered at the upper end of the score range (peak density roughly in the 600-700 range), mostly in the Fair and Good score group. Notably, very few loans were approved for applicants with poor credit scores (below 580). Below this mark, the Poor credit score group faces a steep drop-off in approval rate, reinforcing the structural barrier for conventional

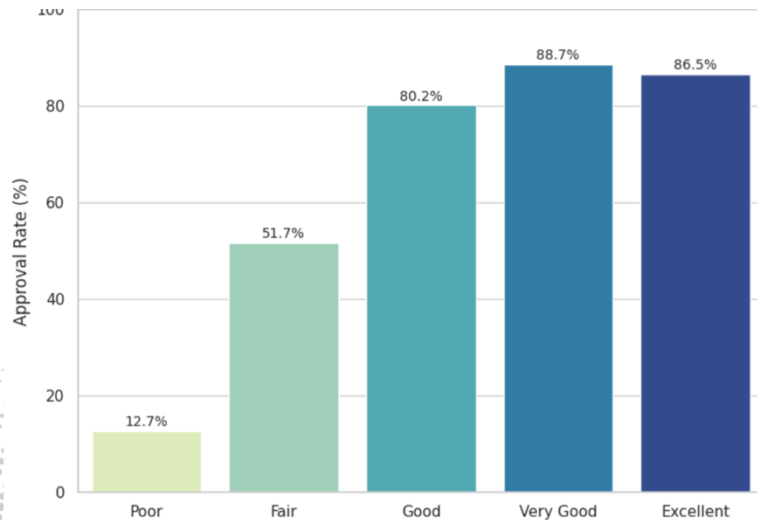


Figure 2.2.2. Loan approval rate by Credit score group

This aligns with the general rule that the higher a borrower's credit score, the more likely they are to be approved. Lenders view a high score as an indicator of lower risk, so those borrowers get the green light more often. More specifically, moving from *Poor* to *Fair* increases your approval chance by more than 40%, reaching 51.7%. And once you hit the *Very Good* tier, your likelihood rises by another 1.5 times.

Credit score can be viewed as a borrower's financial résumé, reflecting historical financial behavior such as credit history length, employment duration, and age. Our analysis shows a clear positive linear relationship between age, years of credit history, and credit score.

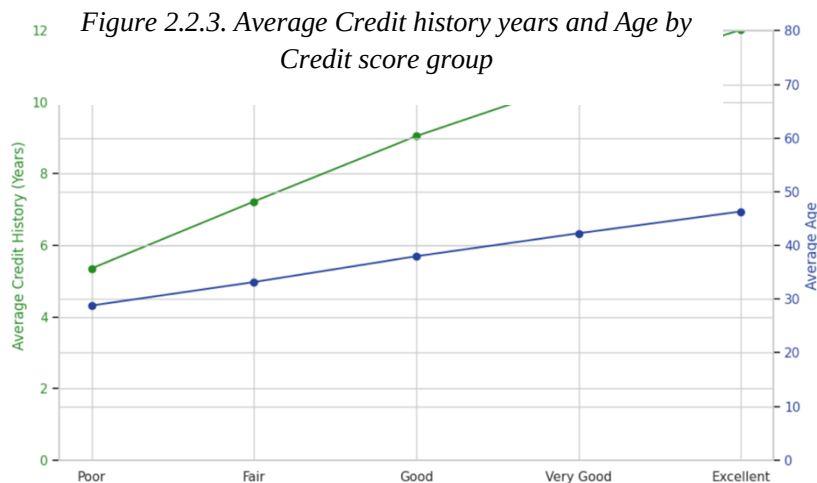


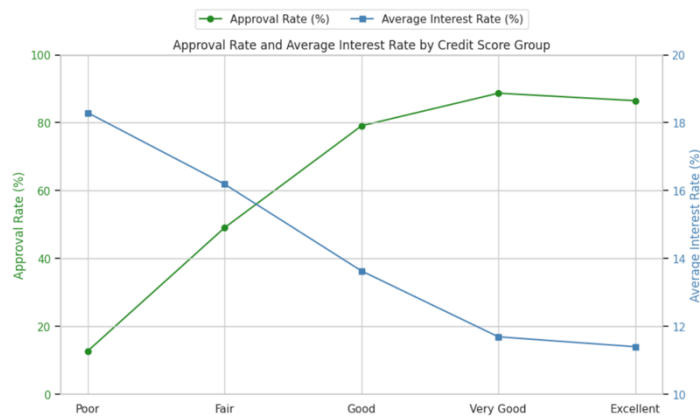
Figure 2.2.3. Average Credit history years and Age by Credit score group

One reason credit score strongly relates to loan outcomes is that it improves with longer and more stable credit experience. In the dataset, lower-score borrowers were generally younger and had shorter credit histories. For instance, the sub-580 group averaged around 29 years old with only 5–6 years of credit history. By contrast, mid- and high-score groups were progressively older and had accumulated substantially longer credit histories.

This proves one thing: Time is essential in building credibility. To reach an excellent credit score (above 800), borrowers need a long period — often over 12 years — to demonstrate consistent and responsible debt management. A longer credit history naturally comes with older age, since older individuals have had more time to accumulate financial management experience and build a solid credit profile.

Benefits of a High credit score

Figure 2.2.4. Average Loan approval rate and Interest rate by Credit score group



In addition to loan approval, having a high credit score also gives customers an advantage in terms of interest rates. As shown, the relationship between credit score and interest rate is strongly inverse. The *Poor* group faces the highest interest rate (18.29%), and the rate decreases significantly through the group score, and then comes to the *Exceptional* group, which enjoys the lowest rate (11.40%). Beyond simply

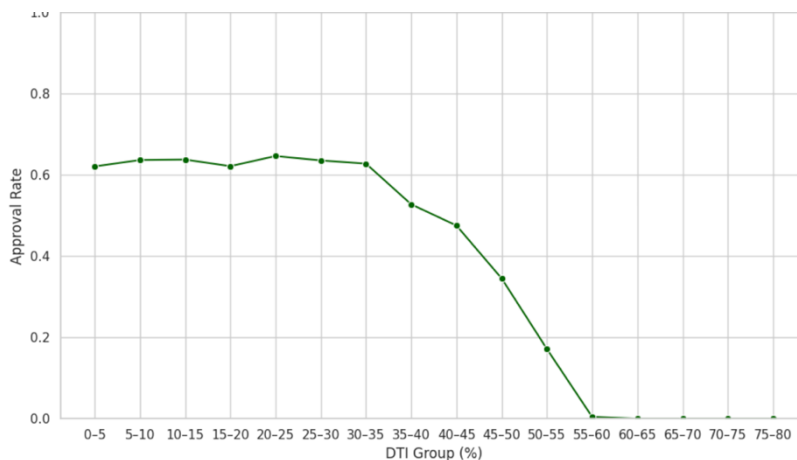
getting approved, a high credit score “rewards” borrowers with better loan terms – notably, lower interest rates. Even though the *Poor* group can still be approved, they must accept much higher interest rates (~18%), allowing banks to compensate for the increased risk. Interest rate, therefore, is a key tool in risk-based pricing.

2.2. DTI (Debt-to-Income) - Capacity to Repay

While the credit score measures the borrower’s history, the DTI ratio measures their present reality. DTI is the percentage of a borrower’s monthly income that goes toward paying debts. A low DTI means the person’s income comfortably exceeds their debt obligations, whereas a high DTI means a borrower has a heavy debt load relative to income.

The “breaking point.”

Figure 2.2.5. Loan approval rate by Debt-to-Income (DTI) group

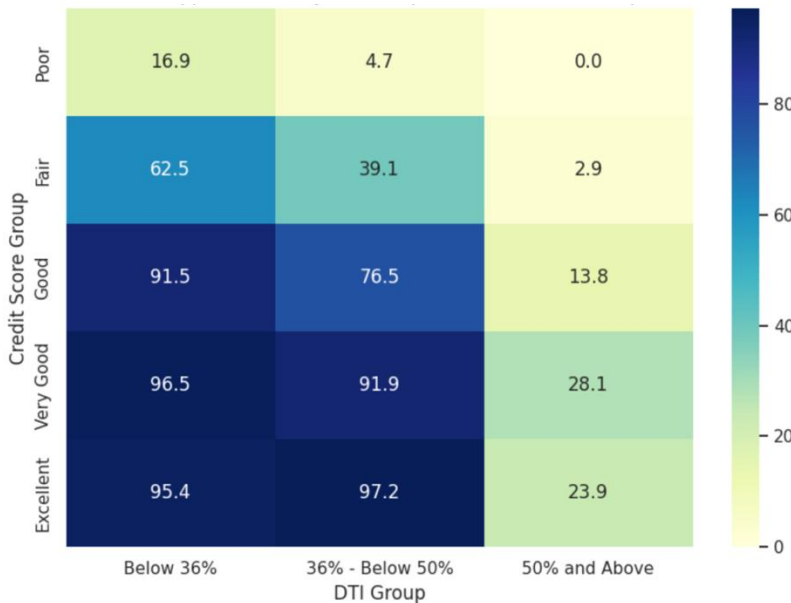


Applicants with a DTI between 10% and 30% show the highest approval counts. The approval count decreases steadily as DTI increases. In our approved-loan data, we found that the majority of approvals fell in the under 36% DTI group. This means the “safe zone” for DTI was generally under ~36%. In fact, if you fall below 35%, you already secure over a 60% approval success rate. The 30–35% DTI range appeared to be a critical

breaking point – beyond ~36%, the majority of loan opportunities were lost to denial. Our analysis reinforces that a debt load above 40% of income is usually a red flag for lenders, dramatically lowering the likelihood of approval.

The power of DTI – Compensating for low Credit scores

Figure 2.2.6. Heatmap of Loan approval rates by Credit Score and DTI group



We analyzed the interaction between DTI and credit score to understand how these two key factors jointly affect loan approval. The results show that a strong credit score can partly offset a higher DTI, but only to a certain extent. In the “Poor” credit score group, applicants with a DTI below 36% still achieved an approval rate of 16.9%— 3.6 times higher than those with medium DTI. This suggests that even with weak credit history, strong current cash flow, reflected by a low DTI, can provide enough confidence for approval. However, extremely high DTI (>50%) significantly reduces approval chances

for all borrowers. Even applicants with excellent credit scores had an approval rate of only 23.9%, although this remained substantially higher than the ~3% approval rate observed among low-score borrowers with similarly high debt burdens.

2.3. Conclusion

As one might say: “Banks aren’t afraid of you borrowing a lot — they’re afraid of you borrowing more than you can afford to repay.” Our data analysis highlights that to maximize loan approval chances, borrowers should focus on both building a trustable credit profile and maintaining a manageable debt load. Credit score reflects your past behavior and trustworthiness as a borrower, while DTI reflects your current financial balance and ability to take on new debt.

3. Lending intent - Elements for Credit risk management

3.1. Overwhelming priority for the Education group

The overall approval rate for Education is consistently the highest, at 66.73%.



Figure 2.3.1 Loan Approval Rate by Annual income group and Loan intent

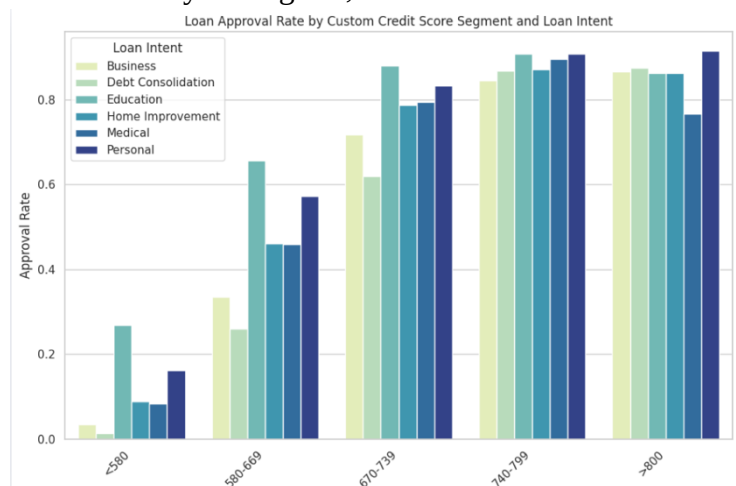


Figure 2.3.2 Loan approval rate by Credit score group and Loan intent

The data shows a consistent trend: Education always has the highest loan approval rate across all groups. Even among customers with low credit scores (Fair–Poor) or low income (below \$25,000), education loans are approved at significantly higher rates than other purposes. While higher income and credit scores improve approval rates for other loan intents, they rarely exceed education, indicating that loan purpose positively influences approval decisions

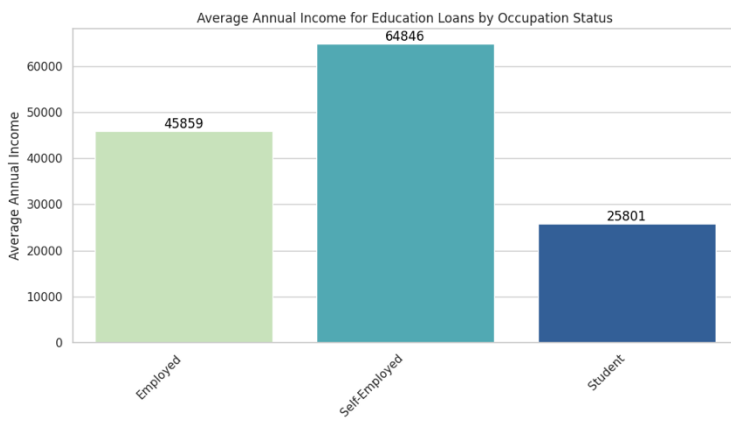


Figure 2.3.3 Average annual income for Education loans by Occupation status

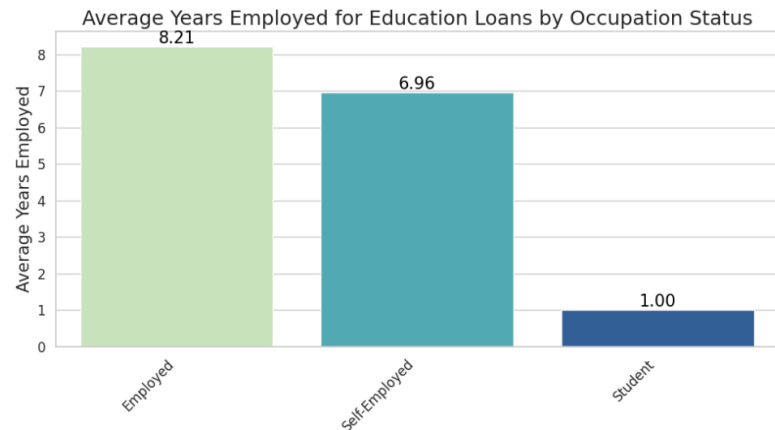


Figure 2.3.4. Average Years Employed for Education Loans by Occupation Status

Across all income levels, Education consistently has the highest approval rate, with the largest gap in the low-income group. Students are strongly prioritized with a 65.86% approval rate due to support policies. In higher-income groups, approval rates remain high (67–68%) because stable employment and the perceived safety of education loans reduce bank risk.

3.2. The paradox between financial capacity and approval rates

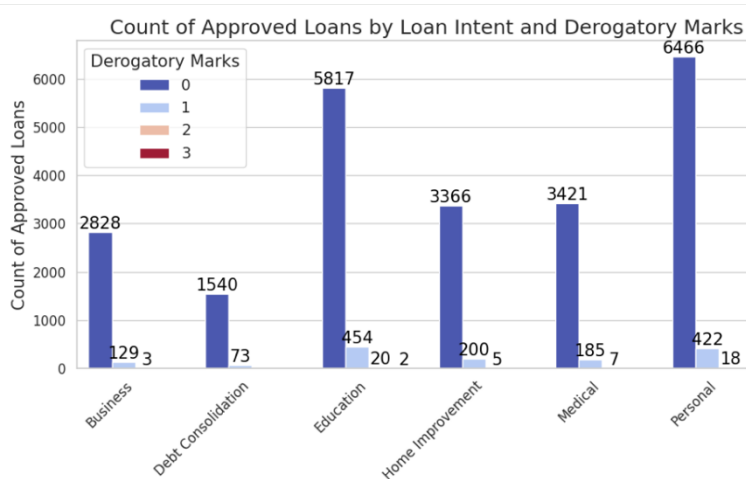


Figure 2.3.5. Count of Approved Loans by Loan Intent and Derogatory Marks

A key finding is the comparison between Education and Business loan profiles. The Business group shows stronger financial indicators, with higher average income (\$57.41k vs. \$52.82k), higher savings (\$5.49k vs. \$3.99k), lower DTI (23.4% vs. 24.65%), and lower delinquency (0.27 vs. 0.36). Despite these advantages, Business loans have a much lower approval rate, confirming a clear approval bias between the two loan intents.

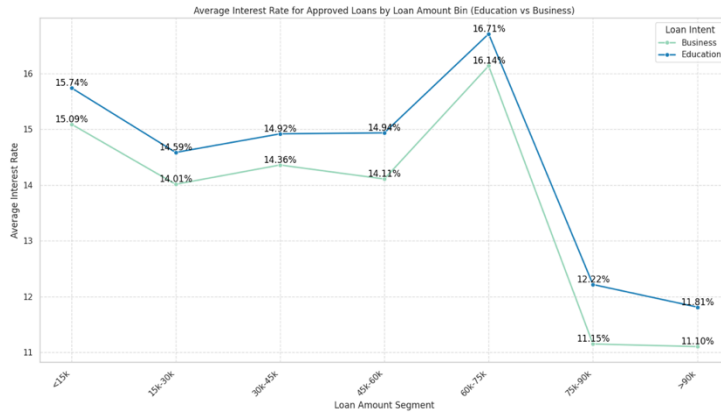
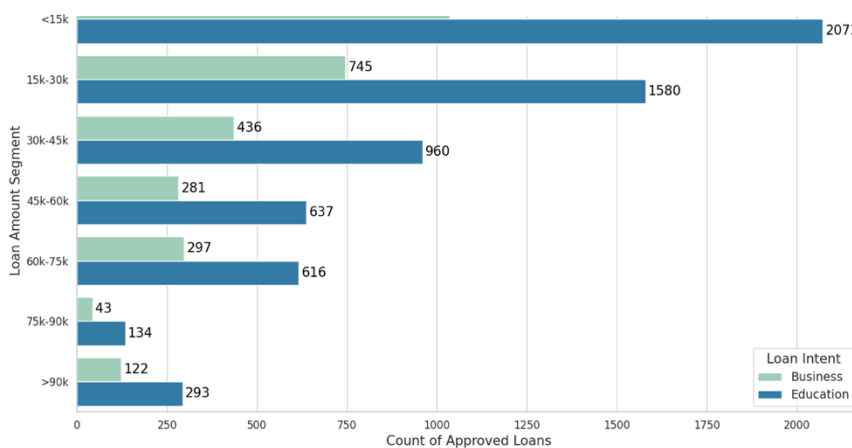


Figure 2.3.6. Average Interest Rate for Approved Loans by Loan Amount Segment for Education vs Business

3.3. Risk pricing strategy and profit optimization

Figure 2.3.7. Count of approved loans by loan amount group for Business vs Education



We can clearly see that the education purpose is prioritized significantly more. The bank applies a risk-based pricing strategy to differentiate Education and Business loans. In most medium loan segments, Education loans carry higher average interest rates than Business loans, reflecting the lender's view that Education loans pose higher repayment risk and therefore require interest rate compensation. Extremely high interest rates (above 16%) are mainly concentrated in the \$60,000–\$75,000 segment, while the lowest rates (around 11%) are observed in loans exceeding \$90,000, driven by strong credit quality. Additionally, the bank's profit strategy relies on high transaction volume, approving more Education loans to accumulate large total loan amounts combined with higher interest rates. Both loan purposes are most concentrated below \$30,000, a popular segment aligned with market demand and optimal risk control, supporting stable cash flow and liquidity.

3.4 Conclusion: The strategy behind each Loan purpose

Education Group (Stable) – Profit Optimization Strategy: Considered safe with social value, the bank loosens standards (e.g., DTI, higher tolerated delinquency) to approve more loans. To offset risk, it applies higher interest rates and relies on loan volume to maximize profit.

Business Group (Higher Risk) – Capital Preservation Strategy: Despite strong-looking profiles, the bank applies the strictest standards due to unpredictable business risks, prioritizing protection of principal capital. As a result, approval rates are consistently lower than for Education.

IV. CONCLUSION

1. Key insights

Loan approval is not determined by a single variable, but by the combined effect of multiple factors. At the same time, banks follow strategic approaches to maximize profit.

A borrower's past behavior remains the most critical "gatekeeper." Applicants with more than 1 default, more than 2 delinquencies, or a DTI exceeding 50% are almost certain to be rejected, regardless of income or asset strength. This confirms that credit history is the prerequisite filter used by banks to decide whether to continue a credit relationship.

While the Credit Score is the strongest influencer because it reflects overall trustworthiness, it works in tandem with the DTI (Debt-to-Income Ratio). These two factors act as a mutual compensation mechanism to control cash flow risk. A high DTI significantly weakens repayment capacity, triggering tighter scrutiny; once it surpasses the 50% threshold, the probability of approval drops to zero, as the cash flow risk outweighs any other positive profile indicators.

Banks also tailor profit strategies based on loan intent. They evaluate not only the borrower but also the risk inherent in the loan purpose. For stable and socially beneficial purposes such as education, the bank is willing to accept higher risk in exchange for a large volume of loans and a higher interest rate to maximize profits. Purposes with inherent risk, like Business, will be strictly screened to mitigate default risk and preserve capital.

Overall, loan approval decisions result from a combination of credit history, repayment capacity, and profit-driven strategies tied to the intent of the loan.

2. Call-to-Action

After gaining deep insights from the dataset, we developed practical strategies to help customers use credit wisely, avoid unnecessary penalties, and increase loan approval chances.

2.1. Borrower profile: Repairing & strengthening credit signals

- Pay all debts on time especially in cases of previous defaults or delinquencies
- Clear overdue loan promptly and update credit records with banks
- Keep credit card utilization below 30% of the total limit
- Provide 3–6 months of bank statements showing stable and regular deposits

2.2. Collateral & Finance: Transparency and Risk mitigation

- With high credit risk (e.g., poor history, high DTI), collateral becomes strongest backup.
- Use assets clear status and complete documentation as collateral
- Ensure collateral is free of disputes or zoning restrictions
- Demonstrate self-financed capital of 30-50% of the loan value
- Show equity through saving accounts, time deposit, or certified bank balances
- Use supplementary assets (vehicles, savings books...) to reduce credit risk

2.3. Documentation & procedure: Completing a High-quality file

- Ensure consistency and accuracy across all submitted documents
- Avoid administrative errors such as missing signatures or incorrect information
- Provide honest and transparent financial information
- Prefer bank transactions over cash to build traceable financial records
- Cooperate closely with credit officers throughout the application process

In summary, loan approval is not just about meeting basic criteria but strategically proving financial discipline. To make these strategies actionable, we have designed a professional brochure titled "**Decoding your Loan Application**". This guide serves as a practical roadmap, helping borrowers quickly grasp the core steps to optimize their applications. Please refer to the attached brochure [here](#) for more details.

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Dataset

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